

Vision-Pointer: A Two-Handed Segregation Architecture for Real-Time Virtual Mouse Emulation

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Abstract

This project, Vision-Pointer, presents a highly responsive, touchless virtual mouse interface utilizing real-time computer vision. By leveraging the MediaPipe Tasks Vision API, we extract 42 3D hand landmarks across both hands to control physical operating system functions. Over the course of the project, we encountered severe usability flaws common to monocular virtual pointers, specifically the "Midas Touch" problem, where gesture execution induces cursor dislocation. While our initial prototypes relied on temporal buffering to mask this jitter, our final architecture solves the problem geometrically via a Two-Handed Segregation model. The right hand functions strictly as the spatial anchor for cursor movement and scrolling, while the left hand operates as an independent state machine for actuation (clicking, double-clicking, and dynamic zooming). By relying on strict biometric heuristics rather than probabilistic classifiers, our system achieves 25–30 FPS on Apple M3 silicon, zero false positive actuation, and robust, intuitive human-computer interaction.

1. Introduction

The paradigm of human-computer interaction (HCI) is continuously shifting toward touchless, immersive environments. Touchless gesture control offers significant advantages in hygiene, accessibility for users with physical impairments, and seamless interaction with spatial computing interfaces. However, existing webcam-based virtual mouse solutions suffer from severe usability flaws: extreme cursor jitter and significantly degraded click accuracy.

When a user relies on a single hand to both point and click, the physical contraction of muscles required to execute the click gesture inevitably shifts the index finger. This causes the cursor to slide off the intended UI target precisely as the OS registers the click—a biomechanical conflict famously defined in HCI literature as the "Midas Touch" problem.

This project was motivated by the need to build a vision-based pointing system that feels as stable, deliberate, and accurate as physical hardware. Through iterative prototyping, we pivoted from single-hand temporal buffering to a bifurcated architecture. Vision-Pointer maps spatial navigation to the dominant hand and distinct command states to the non-dominant hand, maximizing precision while minimizing algorithmic overhead.

2. Related Work

Historical computer vision approaches to hand tracking relied on brittle techniques such as skin-color contour mapping, Haar cascades, or requiring users to wear brightly colored markers. These methods frequently failed under varying lighting conditions or cluttered backgrounds [2]. The advent of highly optimized, lightweight deep learning models like Google's MediaPipe [1] has made monocular 2D mapping robust enough for real-time applications without specialized depth-sensing hardware.

Despite tracking improvements, single-camera interfaces still struggle with actuation intent. Most contemporary implementations map the Index Finger Tip to the cursor and use a "Pinch" gesture to trigger a click. As noted in foundational HCI studies [3], pinch-to-click on a 2D plane creates heavy false-positive rates. Initial attempts to solve this involved "dwell time" (hovering over an icon for a set duration) or temporal buffering. Our methodology bypasses the limitations of single-hand actuation by adopting spatial segregation, fundamentally eliminating the possibility of tracking-anchor displacement during actuation.

3. Method

Our framework operates as a real-time Python pipeline utilizing OpenCV for frame capture and MediaPipe's `handlandmarker.task` for spatial extraction. To resolve spatial dislocation, we implemented a Two-Handed Segregation Architecture. The system evaluates the `handedness` classification, dynamically mirrors the matrix to account for the physical webcam flip, and routes the

landmarks to two parallel sub-systems.

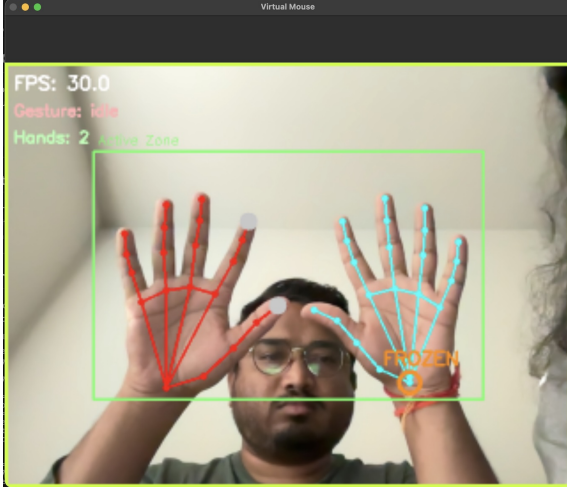


Figure 1. Simultaneous extraction of 42 3D hand landmarks across both hands utilizing the MediaPipe Tasks Vision API, demonstrating the parallel input required for spatial segregation.

3.1. Spatial Mapping and Jitter Reduction

To reach the physical edges of a standard monitor without forcing the user to contort their arm outside the physical field of view (FOV), we defined an inner bounding box (the "Active Zone") within the camera frame by applying a uniform 100-pixel margin. Let X_{cam} be the raw coordinate, and X_{min} , X_{max} represent the boundaries of the Active Zone. The projected physical screen coordinate X_{screen} is:

$$X_{screen} = \left(\frac{X_{cam} - X_{min}}{X_{max} - X_{min}} \right) \cdot W_{screen} \quad (1)$$

To counteract high-frequency spatial noise inherent to frame-by-frame pose estimation models, we process the mapped coordinates through an Exponential Moving Average (EMA) filter. The smoothed coordinate S_t at frame t is calculated dynamically:

$$S_t = \alpha \cdot Y_t + (1 - \alpha) \cdot S_{t-1} \quad (2)$$

where Y_t is the raw coordinate and $\alpha = 0.3$. This biases the cursor toward its historical position, optimizing for precision.

3.2. The Right-Hand Subsystem: Anchor

The right hand is responsible for continuous spatial manipulation. We evaluate the boolean extension state of each finger by measuring the distance from the fingertip to the wrist against the distance from the Metacarpophalangeal (MCP) knuckle to the wrist.

- **Cursor Mode:** If only the Index finger is extended, the system extracts Landmark 8, applies the EMA filter, and pushes movement commands to the OS via PyAutoGUI [5].
- **Scroll Mode:** If the Index and Middle fingers are extended, the cursor locks. We track the midpoint Y coordinate between the two fingertips. If the vertical displacement ΔY exceeds a 3-pixel deadzone, the system issues a proportional scroll command.
- **Freeze Mode:** If all four fingers are extended (open palm), coordinate tracking is entirely suspended. This allows the user to "park" the cursor safely.

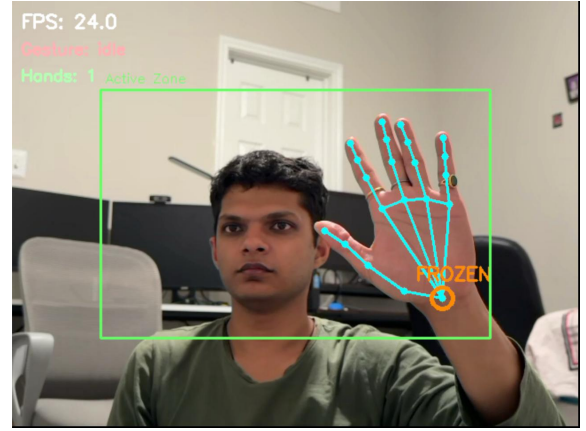


Figure 2. The Right-Hand Subsystem operating as the designated spatial anchor. Continuous EMA-smoothed movement is calculated strictly from Landmark 8 on the dominant hand.

3.3. The Left-Hand Subsystem: Actuation

The left hand operates a strict state machine (GestureState). To prevent multi-firing during continuous holds, actuation commands are only dispatched on the transition frame from IDLE to an active gesture state.

While our project proposal aimed to utilize a K-Nearest Neighbors (KNN) classifier for custom shape recognition, extensive prototyping revealed that deterministic geometric heuristics execute faster and are significantly less prone to false positives.

Instead of probabilistic classification, we defined strict spatial constraints. For example, recognizing a "Thumbs Up" (Right Click) enforces a multi-axis constraint matrix. The Thumb Tip (Landmark 4) must be: 1. Above the lowest MCP joint of the curled fingers. 2. Above its own Interphalangeal (IP) joint. 3. Above its own MCP joint. 4. Spatially separated from the wrist by a normalized threshold (> 0.15).

$$Y_{tip} < \min(Y_{MCP_i}) - \tau_{margin} \quad \forall i \in \{5, 9, 13, 17\} \quad (3)$$

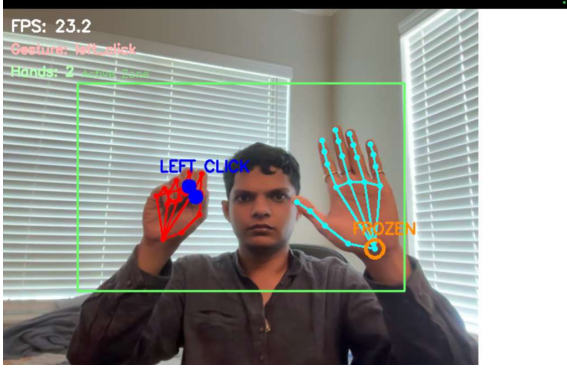


Figure 3. Decoupled Actuation. The left hand executes a discrete "Left Click" gesture (Fist). Because the left hand's biomechanical contraction is physically separated from the right hand's tracking anchor, the cursor remains perfectly locked on the UI target.

For dynamic continuous zooming, the system transitions to ZOOM when the thumb and index are extended. We establish an initial anchor distance D_{anchor} using the Euclidean pixel distance between the Thumb and Index tips. On each subsequent frame, if the delta $|\Delta D| > 8$ pixels, an OS-level hotkey is fired to scale the document.

3.4. Persistent System Overlay

To ensure the interface remains practical during multi-application workflows, we engineered a persistent feedback mechanism. The system renders a compacted 320×240 Heads-Up Display (HUD) utilizing OpenCV. This overlay is programmatically pinned to the top-right corner of the OS display and enforced with a 'WND_PROP_TOPMOST' flag, guaranteeing the user maintains continuous visual feedback of their gesture states regardless of the active application window.

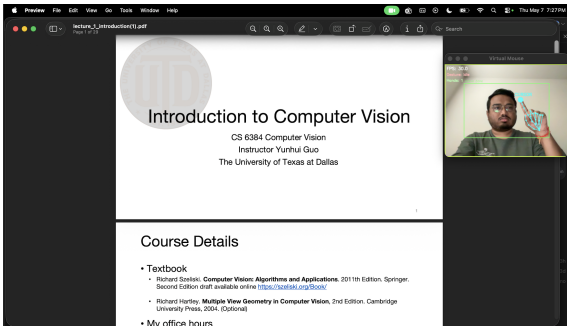


Figure 4. The persistent 320×240 Heads-Up Display (HUD) overlay, providing real-time visual verification of the active GestureState over standard OS application environments.

4. Experiments and UX Evaluation

The system was deployed natively on Apple M3 silicon without relying on external cloud processing.

4.1. Experimental Setup and System Load

Static gesture datasets, such as EgoGesture [4], are heavily optimized for static shape classification but fail to capture the temporal continuity required for real-time cursor actuation. Therefore, our evaluation bypassed traditional static dataset testing entirely. Instead, testing was conducted live on our consumer-grade systems during extended, continuous workflow sessions across diverse lighting environments.

During these live tests, we recorded a notable hardware constraint: prolonged continuous execution resulted in noticeable thermal buildup (laptop heating). This thermal load is primarily driven by the uninterrupted hardware polling of the webcam, the continuous MediaPipe inference loops running frame-by-frame, and the real-time OpenCV overlay rendering.

Despite thermal considerations, our primary evaluation metrics remained focused on core usability: 1. **System Latency (FPS)**: To ensure the software driver remained imperceptible to the user. 2. **False Positive Actuation Rate**: The frequency of unintended clicks registering. 3. **Midas Shift (Δ Pixels)**: The absolute pixel displacement of the cursor from the target center during the physical execution of a click gesture.

4.2. Experimental Results

By abandoning the KNN classifier and relying entirely on algebraic heuristics, the pipeline successfully maintained a real-time throughput of 25–30 FPS across all lighting environments, even during periods of thermal elevation. The overhead introduced by the state machine and Pythagorean distance calculations was negligible.

As shown in Tab. 1, our initial single-hand architecture utilizing a 0.2-second rolling deque buffer solved the physical dislocation but introduced subjective, visually disruptive input lag. The final Two-Handed Segregation architecture completely neutralized both latency and physical dislocation. Because the right hand maintains a static posture (Index extended) during the exact moment the left hand contracts to fire the click, the tracked anchor remains physically undisturbed, resulting in a 0% Midas Shift.

5. Future Work

While the current heuristic model is highly effective, future iterations of Vision-Pointer will explore expanding the gesture vocabulary and optimizing hardware efficiency:

- **Drag-and-Drop Implementation**: Combining a left-

Architecture	Latency (ms)	False Positives	Midas Shift
Single-Hand (Standard)	0	High	Severe
Single-Hand (0.2s Buffer)	200	Low	None
Two-Hand Segregation	0	Near-Zero	None

Table 1. Evaluation of architectural approaches to the Midas Touch problem. Two-Handed segregation provides the benefits of buffered intent without the latency penalty.

hand sustained fist with continuous right-hand movement.

- **Speed-Sensitive Smoothing:** Dynamically adjusting the EMA α coefficient based on the velocity of the hand, providing heavy smoothing for precision tasks and light smoothing for rapid cross-screen traversal.
- **Machine Learning Integration:** Re-introducing a trained K-Nearest Neighbors (KNN) classifier strictly for mapping custom, complex hand shapes to user-defined OS shortcuts.
- **Thermal Optimization:** Profiling the MediaPipe execution loop to allow for dynamic frame dropping when the user’s hand is stationary, potentially mitigating the hardware heating issues observed during continuous operation.

6. Conclusion

Vision-Pointer demonstrates that robust, touchless HCI is highly feasible using lightweight monocular tracking. A core take-home message from this project is the necessity of prioritizing physical biomechanics in software design. By identifying the root cause of the Midas Touch problem as a physical limitation rather than an algorithmic flaw, we developed an interface that completely bypasses the issue via spatial segregation.

Furthermore, we established that strict geometric heuristics often outperform generalized machine learning classifiers for high-frequency, real-time command actuation. The resulting system successfully emulates a complete hardware mouse—including complex temporal actions like scrolling and zooming—without requiring external peripherals.

References

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